

AZURE DEVOPS PROJECT DOCUMENTATION ENABLE AZURE STORAGE ACCOUNT DIAGNOSTIC SETTINGS USING AZURE DEVOPS PIPELINE AND POWERSHELL

Phase 1 — Azure prerequisites (do this first)

Step 1: Verify your existing resources in Azure Portal. Go to Resource Group prasad-rg and confirm all three exist: storage accounts labaccount434 and labaccount435, and Log Analytics Workspace logs. If the workspace is missing, create it via Monitor → Log Analytics Workspaces → Create.

Step 2: Register resource providers. Open Azure Cloud Shell and run these two commands — without this, the diagnostic settings API call will fail with InvalidAuthenticationToken.

```
az provider register --namespace Microsoft.Insights
```

```
az provider register --namespace Microsoft.OperationInsights
```

```
az provider show --namespace Microsoft.Insights --query registrationState
```

Step 3: Create a Service Connection. In Azure DevOps go to Project Settings → Service Connections → New Service Connection → Azure Resource Manager → Service Principal (automatic). Name it exactly as you'll reference it in the YAML (e.g. Azure-ServiceConnection). Grant it Contributor access on the subscription.

Phase 2 — Azure DevOps environments

Step 4: Create the PreProd environment. Go to Pipelines → Environments → New Environment. Name it PreProd, leave resource type as None. No approvals needed here.

Step 5: Create the Prod environment with approval. Create another environment named Prod. Then open it → Approvals and Checks → Add → Approvals → add your sir/team lead as approver → Save. Now any pipeline job targeting environment: Prod will pause and wait for manual approval.

Phase 3 — Repository setup

Step 6: Create the folder structure in your Azure Repos Git repo. You need exactly this layout:

```
/
├── pipeline.yaml
└── scripts/
    └── enable-diagnostics.ps1
```

Phase 4 — The PowerShell script

Step 7: Create scripts/enable-diagnostics.ps1

For the file go through this github link <https://github.com/LALITHAPRASAD272/repo2.git>

Phase 5 — The YAML pipeline

Step 8: Create pipeline.yaml at the repo root

For the file go through this github link <https://github.com/LALITHAPRASAD272/repo2.git>

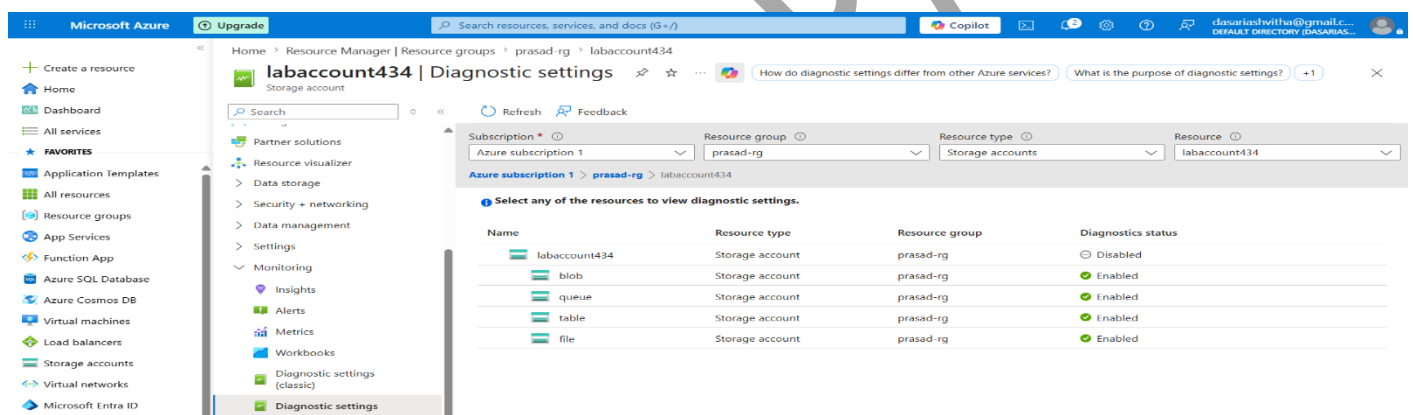
Phase 6 — Create and run the pipeline

Step 9: Register the pipeline. In Azure DevOps go to Pipelines → New Pipeline → Azure Repos Git → select your repo → Existing Azure Pipelines YAML file → select /pipeline.yaml → Save (don't run yet).

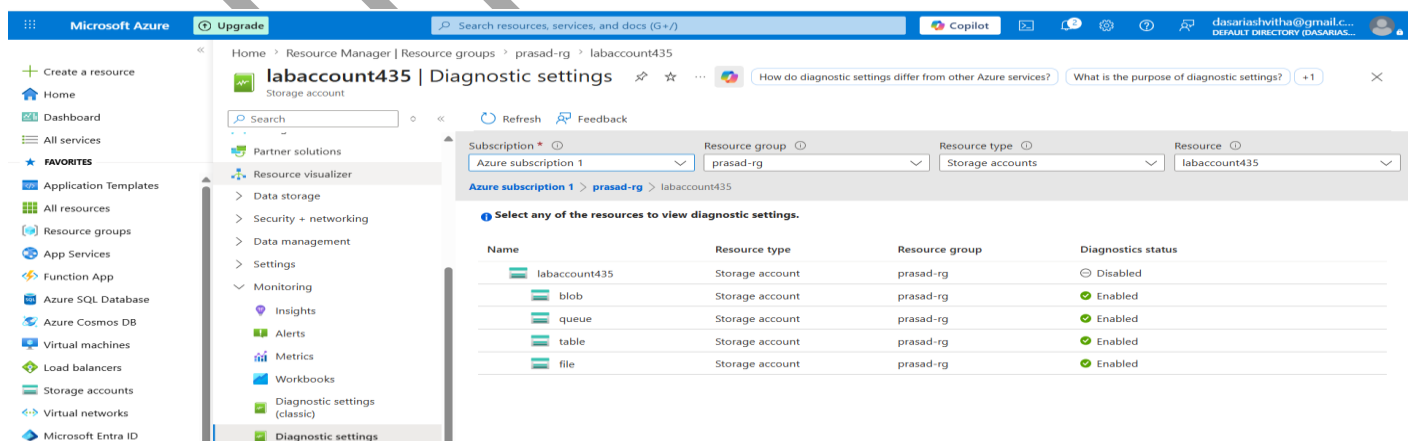
Step 10: Run the pipeline. Click Run Pipeline. Watch the PreProd stage complete first. When it succeeds, the Prod stage will pause and send an approval notification. The approver logs in and clicks Approve. Prod then runs automatically.

Phase 7 — Verification

Step 11: Verify in Azure Portal. Open labaccount434 → Monitoring → Diagnostic settings. You should see four entries: blob-diagnostics, queue-diagnostics, table-diagnostics, file-diagnostics — all pointing to the logs workspace.



Name	Resource type	Resource group	Diagnostics status
labaccount434	Storage account	prasad-rg	Disabled
blob	Storage account	prasad-rg	Enabled
queue	Storage account	prasad-rg	Enabled
table	Storage account	prasad-rg	Enabled
file	Storage account	prasad-rg	Enabled



Name	Resource type	Resource group	Diagnostics status
labaccount435	Storage account	prasad-rg	Disabled
blob	Storage account	prasad-rg	Enabled
queue	Storage account	prasad-rg	Enabled
table	Storage account	prasad-rg	Enabled
file	Storage account	prasad-rg	Enabled

Step 12: Verify logs are flowing. Go to Log Analytics Workspace logs → Logs. Run this KQL query (wait 5–15 minutes after first run for ingestion):

```
kusto
```

```
AzureDiagnostics
```

| where ResourceProvider == "MICROSOFT.STORAGE"

| order by TimeGenerated desc

If you see rows, the full solution is working end-to-end. To trigger log activity, upload a file to a container in one of the storage accounts, then re-run the query.

The screenshot shows the Microsoft Azure portal interface. On the left, the 'logs | Logs' section is selected under 'Log Analytics workspace'. The main area displays a 'New Query 1' window with the following KQL query:

```
1 StorageBlobLogs
2 | order by TimeGenerated desc
3 | take 50
```

The query results are shown in a table with the following columns: TimeGenerated [UTC], AccountName, Location, Protocol, OperationName, and AuthenticationType. The results show several log entries for 'labaccount434' on 'canadacentral' using 'HTTPS' protocol, with 'GetBlobServiceProperties' as the operation name and 'TrustedAccess' as the authentication type.

TimeGenerated [UTC]	AccountName	Location	Protocol	OperationName	AuthenticationType
> 6/6/2026, 11:57:28.096 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess
> 6/6/2026, 11:57:28.068 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess
> 6/6/2026, 11:56:15.304 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess
> 6/6/2026, 11:56:15.134 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess
> 6/6/2026, 11:41:16.959 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess
> 6/6/2026, 11:41:16.396 AM	labaccount434	canadacentral	HTTPS	GetBlobServiceProperties	TrustedAccess

The screenshot shows the Azure DevOps interface for a pipeline named '#20260606.1 • Set up CI with Azure Pipelines'. The pipeline is in a 'Completed' state, indicated by a green checkmark. The summary section shows that the pipeline was executed by 'DASARI ASHVITHA' yesterday at 5:50 PM, taking 34m 38s. The repository is 'repo2.git' and the build is for the 'main' branch, commit '01804c1a'. The pipeline has 0 work items and 0 published items. The stages section shows two jobs, both of which completed successfully with 1 check passed each.

Summary Environments Code Coverage

Individual CI by **DASARI ASHVITHA**
Yesterday at 5:50 PM 34m 38s
Repository and version: repo2.git, main, 01804c1a
Related: 0 work items, 0 published
Tests and coverage: Get started

Stages Jobs

- Enable Diagnostics - ...
1 job completed 1m 37s
1 checks passed
- Enable Diagnostics - ...
1 job completed 1m 21s
1 checks passed